DISSERTATION RESEARCH PROPOSAL

Floor Breach Probability for a GB–France Electricity Interconnector Under Ofgem's Cap and Floor Regime

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1. The Problem

Ofgem's Cap and Floor regime sets a revenue floor below which consumers top up the developer, and a revenue cap above which developers return money to consumers. For Aquind — the proposed 2 GW GB–France interconnector targeting connection in 2027 — these levels are determined by applying Ofgem's Window 3 financial parameters (published December 2024) to Aquind's submitted capital costs and are fixed in real terms for 25 years from Final Investment Decision.

The floor does not eliminate cashflow risk in the way it appears to. When revenues fall below the floor, consumers top up the shortfall — but only at the End of Period Assessment, which occurs every few years. In the intervening period the developer must service its debt from whatever revenues it earns. The frequency and depth of below-floor years is therefore the central financial risk for both the developer and its lenders.

Ofgem's existing revenue assessment does not answer this question. Arup's Market Modelling Report (March 2024), which underpins Window 3 decisions, models three Future Energy Scenarios — Leading the Way, Consumer Transformation, and Falling Short — each under three weather years. Each combination produces a single annual revenue figure. No probabilities are attached to any outcome. For Aquind, Arup concluded that floor payments are unlikely to be required in any scenario. That conclusion rests on nine deterministic runs, not on a statistical model of how the GB–France price spread behaves.

> **Core gap: No probabilistic estimate of floor breach frequency exists for any Window 3 interconnector. This dissertation fills that gap for Aquind using five years of post-Brexit Bloomberg hourly price data and a calibrated stochastic model of the GB–France price spread.**

2. Research Question

Given a calibrated statistical model of the GB–France price spread, what is the probability that Aquind's annual revenues fall below Ofgem's floor in each year of the 25-year regime — and how does cannibalisation from Window 3 build-out change that probability?

As a direct output of the same simulation, the analysis also produces the P90 annual revenue in each year — the revenue level Aquind exceeds with 90% probability. This is the number lenders use to size debt and assess whether the project can service its obligations in a bad year without breaching loan covenants.

3. Why the Spread Is the Right Focus

Aquind earns congestion rent by flowing electricity from the cheaper market to the more expensive one and capturing the price difference. Annual revenue is approximately: capacity × availability × average hours of positive spread × average spread magnitude. Capacity (2 GW) and availability (assumed 95%) are fixed parameters. The regime floor and cap are fixed at Final Investment Decision. The GB-France hourly price spread is the only stochastic variable. Modelling it well is the entire methodological problem.

The post-Brexit GB-France spread exhibits three statistical properties that any adequate model must capture:

- **Mean reversion:** Extreme spreads compress over hours and days as markets re-equilibrate. Confirmed by Augmented Dickey-Fuller tests on post-Brexit data: the spread is stationary and mean-reverting, not a random walk. This is the empirical basis for the Ornstein-Uhlenbeck process used in the spread model.
- **Volatility clustering:** Periods of high spread volatility follow periods of high volatility. Sapio and Spagnolo (2020) document GARCH persistence coefficients of 0.93 in comparable electricity markets, confirming that time-varying volatility is a structural feature.
- **Jump behaviour:** Discrete price spikes from supply or demand shocks — unexpected nuclear outages, the 2021–22 gas crisis — that smooth diffusion models miss entirely. Cartea and González-Pedraz (2012) demonstrate that jump components improve in-sample model fit by 20-48% on European interconnector spread data.

Note on transferable risks: This analysis focuses on quantifiable market price risk, which is the principal driver of floor breach probability. Regulatory risk (changes to cap and floor parameters, policy reversal), construction risk (delay to Final Investment Decision), and interconnection availability risk are not modelled stochastically. Where data and methodology permit, and with the client's assistance in specifying the relevant parameters, qualitative or scenario-based treatment of these risks may be incorporated as an extension.

4. Methodology

The methodology adapts two peer-reviewed frameworks to the specific regulatory and market context of Aquind's Cap and Floor assessment. Each step below identifies the source paper, what it contributes, and how it is applied here.

4.1 Spread Model: Three-Component Structure

Adapted from: *Cartea, A. and González-Pedraz, C. (2012). 'How much should we pay for interconnecting electricity markets? A real options approach'. Energy Economics, 34(1), pp.14–30.*

Cartea and González-Pedraz (2012) developed a three-component model for electricity price spreads between interconnected markets, calibrated to European day-ahead data. Their paper was written to value interconnector leases as real options; this dissertation borrows the statistical model of spread behaviour only, not the option pricing machinery. The model decomposes the spread into:

- **A deterministic seasonal component f(t) —** estimated by OLS regression of the Bloomberg spread on monthly dummy variables and a linear time trend. This captures predictable seasonal patterns (winter peaks, summer troughs) and the secular drift in the GB-France differential. The long-run mean is adjusted for the post-2037 period, where Arup's Future Energy Scenarios project a structural reversal in the spread direction as GB offshore wind capacity surges.
- **A mean-reverting Ornstein-Uhlenbeck process X(t) —** capturing continuous, smooth fluctuations around the seasonal mean. Estimated by AR(1) regression on the jump-cleaned detrended spread. The regression coefficient gives the mean-reversion speed.
- **A compound Poisson jump process Y(t) —** capturing discrete shocks from events such as nuclear outages or sudden demand surges. Positive and negative jumps are estimated separately, with arrival rates fitted by season and jump sizes fitted to exponential distributions. Jumps are detected using Cartea and González-Pedraz's (2012) iterative algorithm, which distinguishes genuine dislocations from rapid mean reversion by testing whether the new price level falls outside the confidence interval of f(t). This distinction matters because misclassifying mean reversion as jumps artificially inflates jump frequency and distorts tail probabilities.

A restricted ARIMA-GARCH baseline (no jump component) is estimated alongside the full model. Formal model selection using the Akaike and Bayesian Information Criteria determines whether the jump component is warranted by the post-Brexit data. The data selects the model; the choice is not imposed.

4.2 Volatility Clustering: GARCH Extension

Adapted from: *Sapio, A. and Spagnolo, N. (2020). 'The effect of a new power cable on energy prices volatility spillovers'. Energy Policy, 144, p.111488.*

Cartea and González-Pedraz (2012) use a seasonally varying but otherwise constant volatility parameter. Sapio and Spagnolo (2020) estimated VAR-GARCH models on Italian zonal price data around the introduction of the SAPEI interconnector and found GARCH persistence coefficients of approximately 0.93 — meaning volatility shocks die out very slowly. A volatile week in electricity markets tends to be followed by another volatile week. This is directly relevant to floor breach probability: without GARCH, the simulation underestimates the probability of sustained runs of below-floor revenues, because it misses the prolonged high-volatility episodes (the gas crisis of late 2021, the French nuclear crisis of 2022) that most threaten cashflow. A univariate GARCH(1,1) layer is added to the OU residuals to correct for this.

Sapio and Spagnolo (2020) also provide empirical evidence that new cable introductions compress price spreads between interconnected zones as arbitrage becomes more efficient — directly validating the cannibalisation adjustment logic described below.

4.3 Cannibalisation Adjustment

Source: *Arup Group (2024). Market Modelling Analysis for Cap and Floor W3 and Offshore Hybrid Assets. March 2024. Sections 2.2 and 3.2.2.3.*

As more interconnector capacity connects on the GB-France and adjacent borders, price spreads compress because arbitrage becomes more efficient. Arup's Market Modelling Report (2024) quantifies this directly through two scenarios: First Additional (FA), in which Aquind connects as the only new project and captures the highest revenues; and Marginal Additional (MA), in which Aquind connects alongside all other Window 3 candidates and revenues are lower. The revenue gap between FA and MA, per gigawatt of additional border capacity, is used as a cannibalisation coefficient. This is applied year-by-year to the simulated spread paths to determine when each additional gigawatt connects. The simulation produces two revenue distributions per year: FA as the upper bound and MA as the lower bound.

4.4 Monte Carlo Simulation and Outputs

10,000 spread paths are simulated over the 25-year operating life using the calibrated parameters. At each daily time step, the model draws a GARCH-conditional Gaussian shock, updates the OU component, and draws whether a positive or negative jump arrives from the relevant seasonal Poisson process. The full spread is reconstructed as the sum of the three components. Each path produces an annual revenue figure based on Aquind's 2 GW capacity and 95% availability.

In each year, the fraction of 10,000 paths falling below Ofgem's published floor level is the floor breach probability for that year. The P90 annual revenue — the revenue exceeded by 9,000 of the 10,000 paths — is produced as a direct output of the same simulation. The Debt Service Coverage Ratio at P90 is calculated against a representative financing structure. Both FA and MA cannibalisation assumptions are applied, giving a range from best-case to worst-case build-out.

Note: Path count is determined by the standard error of a Monte Carlo probability estimate: $SE = \sqrt{p(1-p)/N}$. At breach probabilities of 5-20%, $N = 10{,}000$ yields $SE \approx \pm 0.3$ percentage points, sufficient for results reported to one decimal place. Convergence is confirmed by running at 1,000, 2,500, 5,000 and 10,000 paths and verifying the estimate stabilises. For rare-event years where $p < 2\%$, $N = 10{,}000$ gives a relative error exceeding 30%, so 50,000 paths are used instead.

5. Data

All data required for the analysis is confirmed available. The table below sets out each input, its source, its role in the model, and its availability status.

Dataset	Source & Series	Role in Model	Status
GB hourly day-ahead prices	Bloomberg Terminal N2EXEH01-24 Jan 2021–Mar 2026	Primary input for spread construction and all model estimation	Downloaded ✓
FR hourly day-ahead prices	Bloomberg Terminal PWNXFR01-24 Jan 2021–Mar 2026	Primary input for spread construction and all model estimation	Downloaded ✓
GBP/EUR exchange rate	Bloomberg Terminal EURGBP Curncy Matching period	Convert French prices to GBP/MWh for spread construction	Downloaded ✓
Aquind FA and MA revenues	Arup Market Modelling Report, March 2024 (Ofgem consultation portal)	Source of cannibalisation coefficient (FA–MA gap per GW)	Publicly available ✓
Ofgem Window 3 floor and cap levels	Ofgem Decision, December 2024	Floor threshold against which breach probability is calculated	Publicly available ✓
Window 3 connection schedule	ENTSO-E TYNDP 2022	Timing of cannibalisation deflation year-by-year	Publicly available ✓

The post-Brexit sample period (September 2021–March 2026; Jan-Sept'21 data unavailable) is selected deliberately. Pre-Brexit GB–France spread behaviour was governed by EU market coupling, under which GB and French prices were jointly optimised. Post-Brexit, GB participates in European day-ahead markets through explicit capacity auctions only, producing a structurally different spread regime. Using pre-Brexit data would contaminate the model with a market structure that no longer applies.

6. Expected Outputs

Three outputs, none of which currently exist in published form for any Window 3 asset:

- **Calibrated stochastic spread model:** Full parameter estimates for the GB–France post-Brexit spread: mean-reversion speed, seasonal volatility, GARCH persistence coefficients, and jump arrival rates and sizes by season. Formal model selection results comparing the jump model against the ARIMA-GARCH baseline. The first transparent, replicable calibration of this spread on post-Brexit data.
- **Floor breach probability year-by-year:** Probability that Aquind's annual revenues fall below Ofgem's floor in each of the 25 regime years, under both FA and MA cannibalisation assumptions. The first probabilistic floor breach estimates for any Window 3 interconnector.
- **P90 revenue and DSCR:** The annual revenue level exceeded with 90% probability in each regime year, and the Debt Service Coverage Ratio at that level against a representative

financing structure. The bankable number lenders require for debt sizing, currently absent from all published Ofgem and Arup documentation.

7. Relevance to Vallorii

The floor breach probability answers a question Ofgem's deterministic methodology cannot answer. The P90 revenue fills the gap in every project finance model for a Window 3 asset. Both outputs are directly usable by developers preparing for Final Project Assessment and by lenders conducting due diligence on Aquind or comparable assets.

The methodology — a calibrated stochastic spread model applied to a specific regulatory revenue floor — is transferable to other border pairs and future Cap and Floor windows. It is a replicable, data-driven workflow with full parameter transparency and auditable source data, consistent with Vallorii's objective of producing analysis that can be independently verified and commercially applied.

7.1 How Vallorii's Involvement Would Help

This is a dissertation project, and the analysis is the student's independent academic work. However, the research engages directly with commercial and regulatory questions where practitioner input adds material value. Specifically:

- **Methodological sense-checking:** Vallorii's familiarity with how asset revenues are modelled in practice helps identify whether the cannibalisation coefficient and ENTSO-E build schedule assumptions reflect current commercial expectations, or whether alternative assumptions are more defensible.
- **Regulatory parameter validation:** The Ofgem floor levels and FA/MA revenue figures used in the simulation are drawn from published documents. Vallorii can flag if there are more recent or more precise figures available through engagement with developers or Ofgem.
- **Output framing:** The P90 revenue and DSCR outputs are most useful if framed in the context of how lenders and developers use these numbers. Vallorii's knowledge of financing structures for comparable assets helps ensure the outputs are presented in a form that is immediately applicable rather than academically self-contained.
- **Industry context:** Where the dissertation draws on Arup's modelling assumptions, Vallorii can help contextualise those assumptions against observed market behaviour, particularly for the post-2037 spread reversal projected under Leading the Way.

The degree and form of engagement is entirely at Vallorii's discretion. The analysis does not depend on proprietary data or confidential inputs.

8. Administrative Requirements

This section sets out the practical arrangements for the collaboration, all of which are proposed in the spirit of keeping the process straightforward for both parties.

Item	Position
Nature of collaboration	This is an independent dissertation project undertaken by a postgraduate student. Vallorii's role is that of industry partner, providing context and light-touch input where relevant. The student retains full academic responsibility for the analysis and conclusions.
Paid engagement	Whether this engagement constitutes a paid opportunity is entirely at the client's discretion. It can function equally well as an unpaid academic collaboration. Either arrangement is workable and does not affect the analysis.

Item	Position
MOU and NDA	A Memorandum of Understanding and/or Non-Disclosure Agreement are optional and at the mutual discretion of both parties. Given that all data used is publicly available and the outputs are for an academic dissertation, formal legal agreements are not required but can be put in place if the client prefers.
Deliverables	The primary deliverable is the dissertation itself, which will include a working model and a full written report. If there is a specific additional deliverable — for example, a standalone article or a presentation — this can be agreed with the client and incorporated into the timeline, subject to mutual agreement on scope and deadline.
Methodology	The methodology set out in Section 4 reflects the best academic framework identified from the literature. If Vallorii considers an alternative approach more practical or more commercially relevant for the interconnector context, this can be discussed at the outset and a mutually agreed methodology adopted. The goal is analysis that is both academically rigorous and industry applicable.
In-person presentation	The student will have full availability during the May–August working period and is flexible on format. Findings can be presented in person at Vallorii's offices, or remotely, or via written report only — whichever is most useful to the client.

9. Project Timeline

The dissertation runs from May 2026 (official start) to 13 August 2026 (final submission deadline). Preparatory work undertaken independently in April 2026 is noted separately. Full-time working availability runs from 1 May to 30 July 2026; the final two weeks of July and early August are reserved for write-up, refinement, and presentation of findings.

Period	Phase	Activities	Mode
April 2026 (pre-start)	Preparation	Finalise data collection: Bloomberg series downloaded, Arup and Ofgem documents compiled. Complete literature review of Cartea & González-Pedraz (2012) and Sapio & Spagnolo (2020). Scope and agree methodology with Vallorii.	Independent
April 2026 (pre-start)	Preparation	Agree administrative arrangements with Vallorii: deliverables, format of engagement, any additional data sources.	Collaborative
May 2026 (Weeks 1–2)	Model build	Implement spread model in Python: seasonal component estimation, jump detection algorithm, OU and GARCH parameter estimation. Initial calibration run on Bloomberg data.	Independent
May 2026 (Weeks 3–4)	Model build	Cannibalisation adjustment implementation. Integrate ENTSO-E build schedule and Arup FA/MA coefficient. Validate model against descriptive statistics.	Independent + check-in with Vallorii
June 2026 (Weeks 1–2)	Simulation	Monte Carlo simulation: 10,000 paths over 25 years. Generate floor breach probability	Independent

Period	Phase	Activities	Mode
		distribution and P90 revenue series under FA and MA.	
June 2026 (Weeks 3–4)	Analysis	DSCR calculation. Sensitivity analysis on key parameters (jump frequency, GARCH persistence, cannibalisation coefficient). Draft results chapter.	Independent + Vallorii sense-check on outputs
July 2026 (Weeks 1–2)	Write-up	Draft dissertation: introduction, literature review, methodology, results. Internal review and revision.	Independent
July 2026 (Weeks 3–4)	Presentation	Finalise findings. Prepare presentation for Vallorii if required. Circulate draft conclusions for any final input.	Collaborative — present to Vallorii
1–13 August 2026	Submission	Final dissertation refinement, formatting, and submission by 13 August deadline.	Independent

Independent tasks in April are preparatory work undertaken before the official dissertation start date. Collaborative tasks (shaded) involve check-ins or input from Vallorii; the degree of involvement at each stage is at the client's discretion.

Key References

Arup Group (2024). Market Modelling Analysis for Cap and Floor W3 and Offshore Hybrid Assets. March 2024. Available via Ofgem consultation portal.

Cartea, A. and González-Pedraz, C. (2012). 'How much should we pay for interconnecting electricity markets? A real options approach'. Energy Economics, 34(1), pp.14–30.

González-Pedraz, C. and Moreno, M. (2021). 'Evaluation of a cross-border electricity interconnection'. Energy, 233, p.121092.

National Grid ESO (2022). Future Energy Scenarios 2022. Warwick: National Grid ESO.

Ofgem (2024). Decision on changes to the financial parameters of the cap and floor regime for Window 3 electricity interconnectors. December 2024. London: Ofgem.

Ofgem (2024). Further consultation on the cap rate for the cap and floor regime for Window 3 electricity interconnectors. July 2024. London: Ofgem.

Sapio, A. and Spagnolo, N. (2020). 'The effect of a new power cable on energy prices volatility spillovers. Energy Policy, 144, p.111488.

ENTSO-E (2022). Ten-Year Network Development Plan 2022. Brussels: ENTSO-E.